VICTORY 2027

Natural Vegetation

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

1. Importance of Forests
2. Types of Vegetation (tropical evergreen)
3.
4.

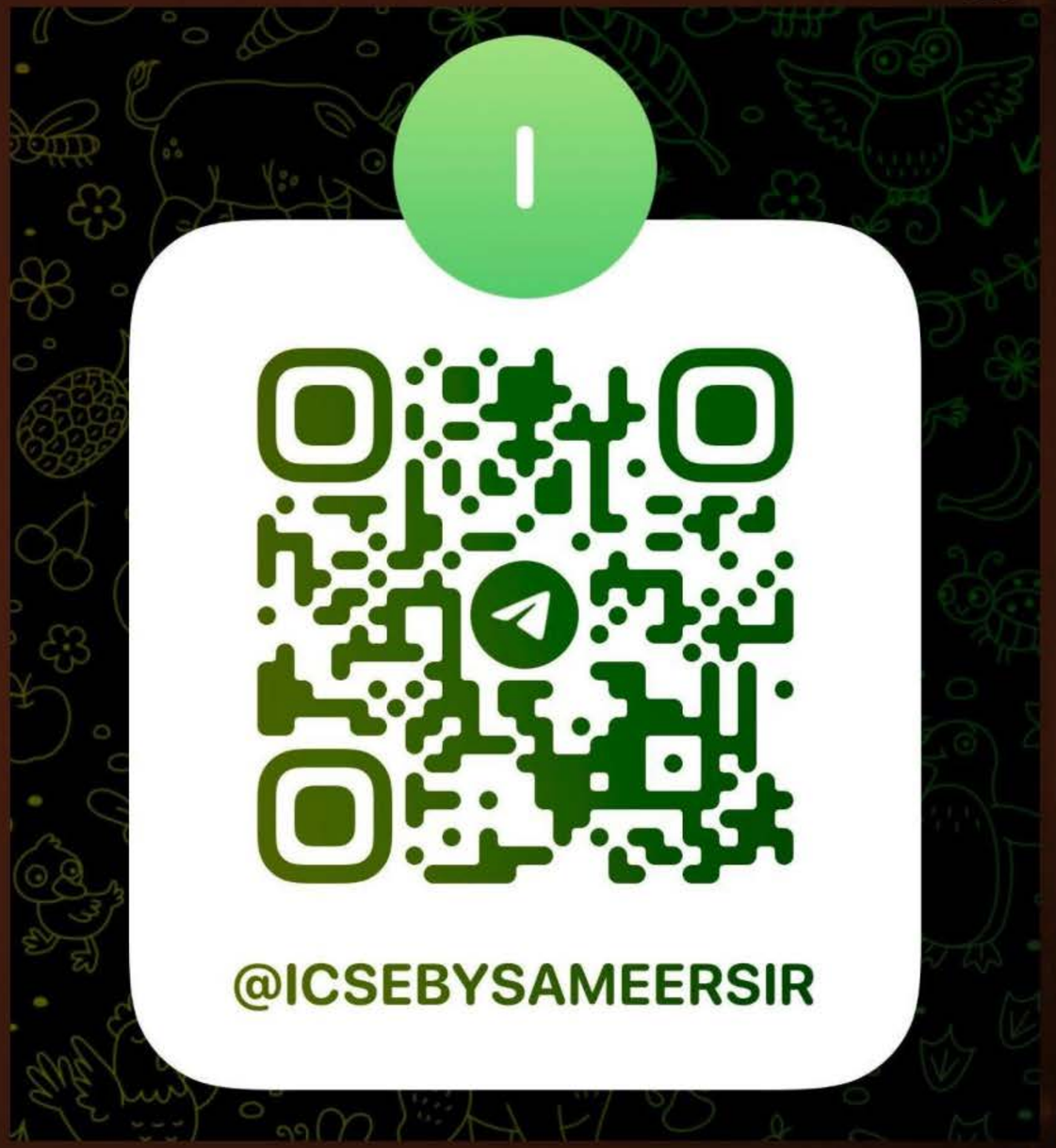
Natural Vegetation

- *Importance of forests.*
- *Types of vegetation; distribution and correlation with their environment (tropical evergreen, tropical deciduous, tropical desert, littoral and mountain).*
- *Forest conservation – afforestation, reforestation, social forestry, agroforestry and Van Mahotsav*



Telegram

https://t.me/icsebysameersir



PYQ's → Soil

Q.1. What is Soil Erosion? → (1)

Q.2 Name any two agents of Soil Erosion? → (1)

Q.3 Suggest Some ways to prevent soil Erosion. → (2)

→ with reference to Black Soil in India answer the following:

i) How is black soil formed?

ii) How is black soil able to retain moisture for a longer period of time.

iii) Mention any two minerals found in black soil.

Q.1 Name the parent material of red soil. → Crystalline metamorphic/ ingneous.

Q.2 How does the soil get it's red Colour.

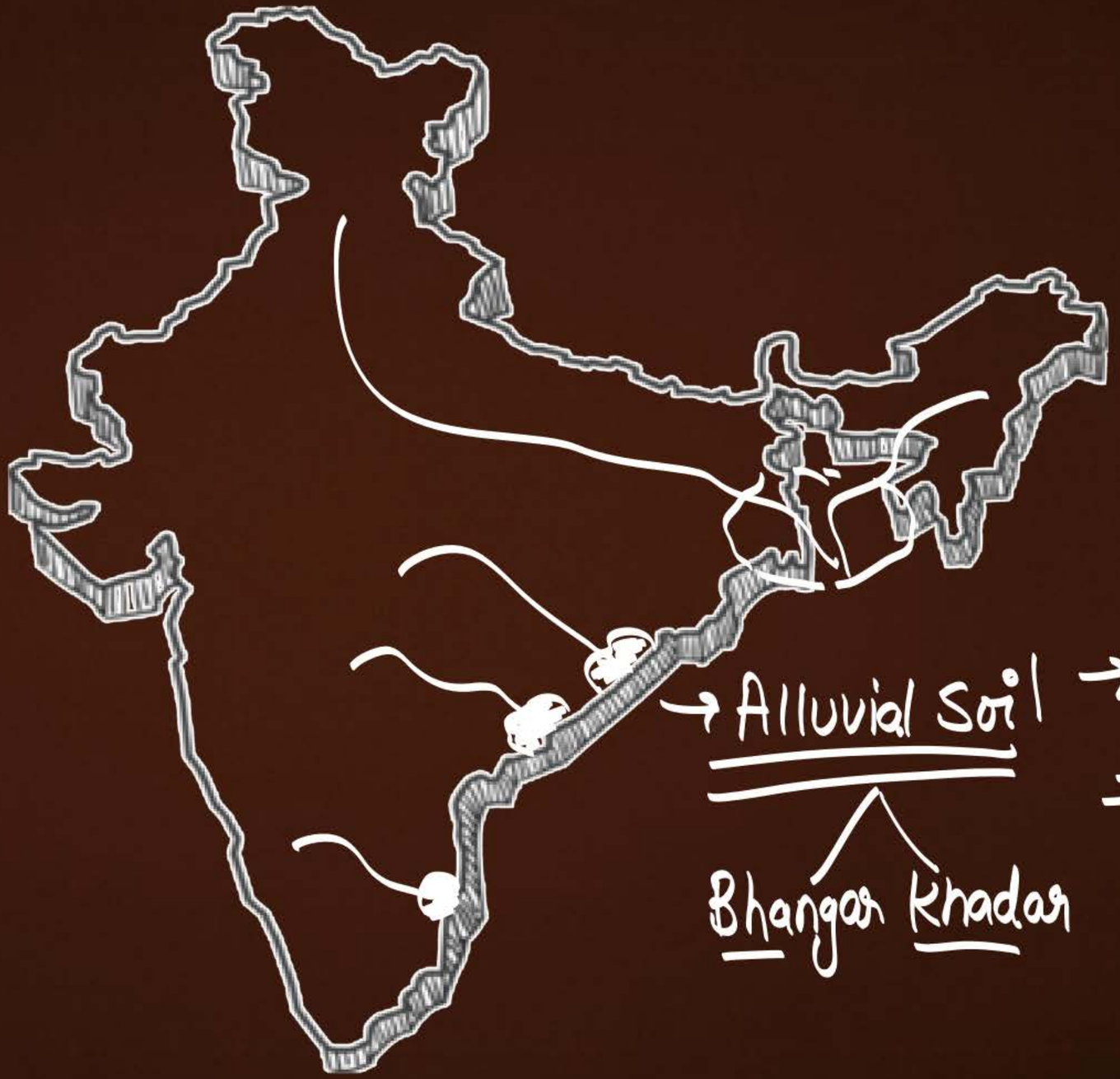
$Fe + O_2 \rightarrow$ Ferrous oxide

Iron oxide → rust → Brown/red.

Yellow Soil

↳ Soil is formed in presence of water.

Iron oxide + water → Yellow


→ Alluvial Soil → River Deposition
Bhangar Khadar

Regur Soil
Cotton Soil
Self-ploughing Soil



- Climatic Variation
- Soil

Cactus




Natural
Vegetation

Natural Vegetation

- Natural Vegetation refers to the plant community which has grown naturally without any human assistance and has not been disturbed over a long time.
- Thus, grasses, shrubs and trees, which grow on their own without any human interference, constitute the natural vegetation of an area.

Natural Vegetation

- There is a difference between flora, vegetation and forest.
- Flora refers to plants of a particular region or period, listed as species and considered as a group.
- For example, the Eastern Himalayas have about 4000 species of plants which vary with increasing altitude from tropical to temperate and alpine.



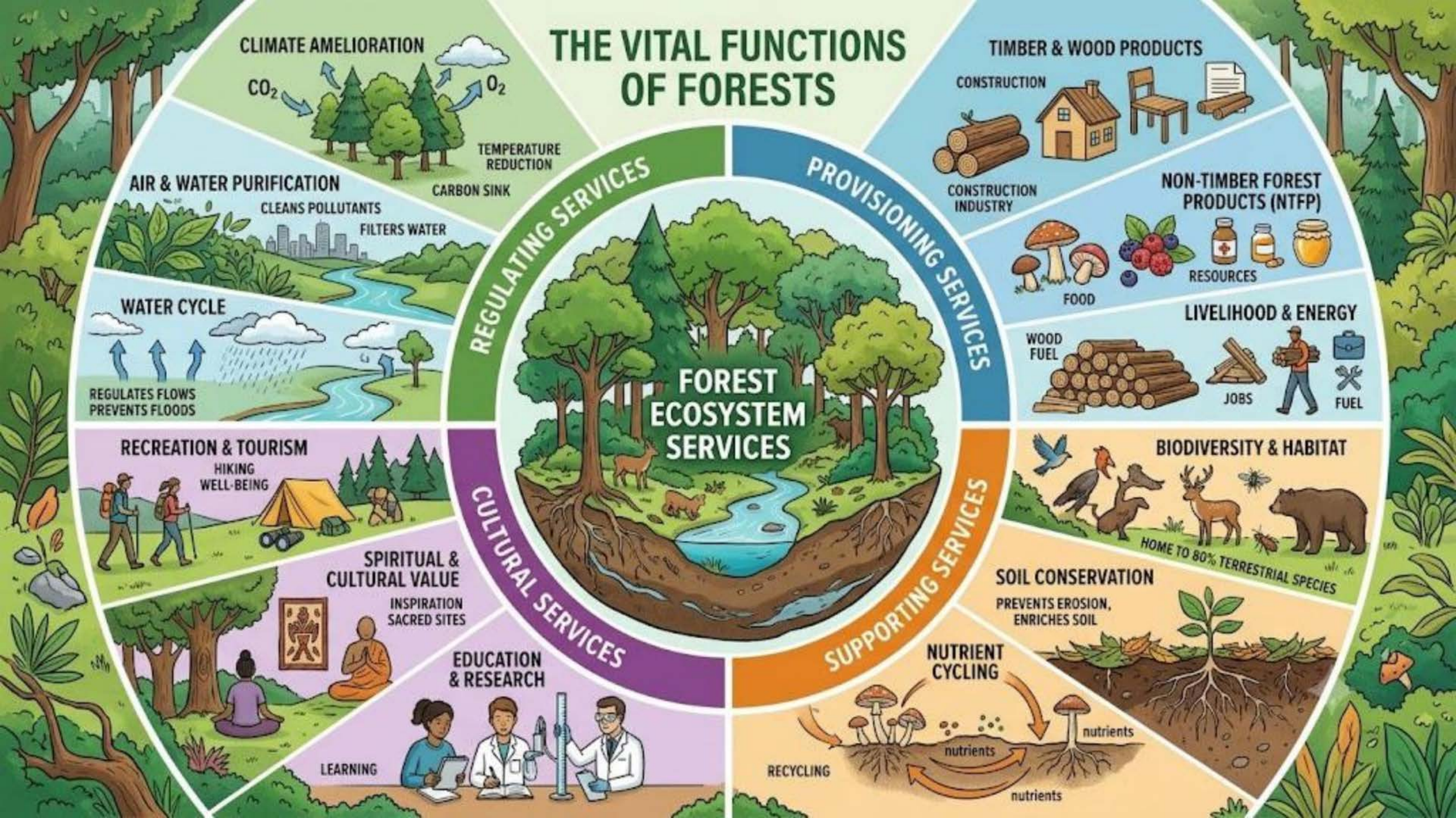
Natural Vegetation

- Vegetation, on the other hand, refers to the assemblage of plant species living in association with each other in a given environmental setup.
- For example, the redwood forests, coastal mangroves, roadside weed patches, cultivated gardens and lawns, etc., all are encompassed by the term vegetation.
- The major vegetation types of the world are grouped as forests, grasslands, scrubs and tundra.

Natural Vegetation

- Forest refers to a large tract of land covered with trees and accompanying undergrowth of shrubs, herbs and sustaining thousands of life forms, which include both plants and animals.

THE VITAL FUNCTIONS OF FORESTS



Importance of Forests



(i) **Productive Functions:** Various trees provide us with products such as fruits, leaves, roots and tubers of plants.

- Wood is used for making furniture in houses as well as industrial units.
- Wood and bamboo pulp are used for manufacturing paper and paper boards.
- Wood is used as a source of energy for cooking and for providing warmth.


WILD EDIBLES
MEDICINAL HERBS
TREE PRODUCTS
WILD EDIBLES
TRUFFLES
FOREST HONEY

Importance of Forests

- Forest products, other than timber and firewood, include fibres, essential oils, oil seeds and edible plants.
- Bamboos provide a means of livelihood for the tribals who make mats, baskets, ropes, etc., using bamboo.
- It is also used in the manufacture of rayon (yarns and artificial silk fibres).

Importance of Forests

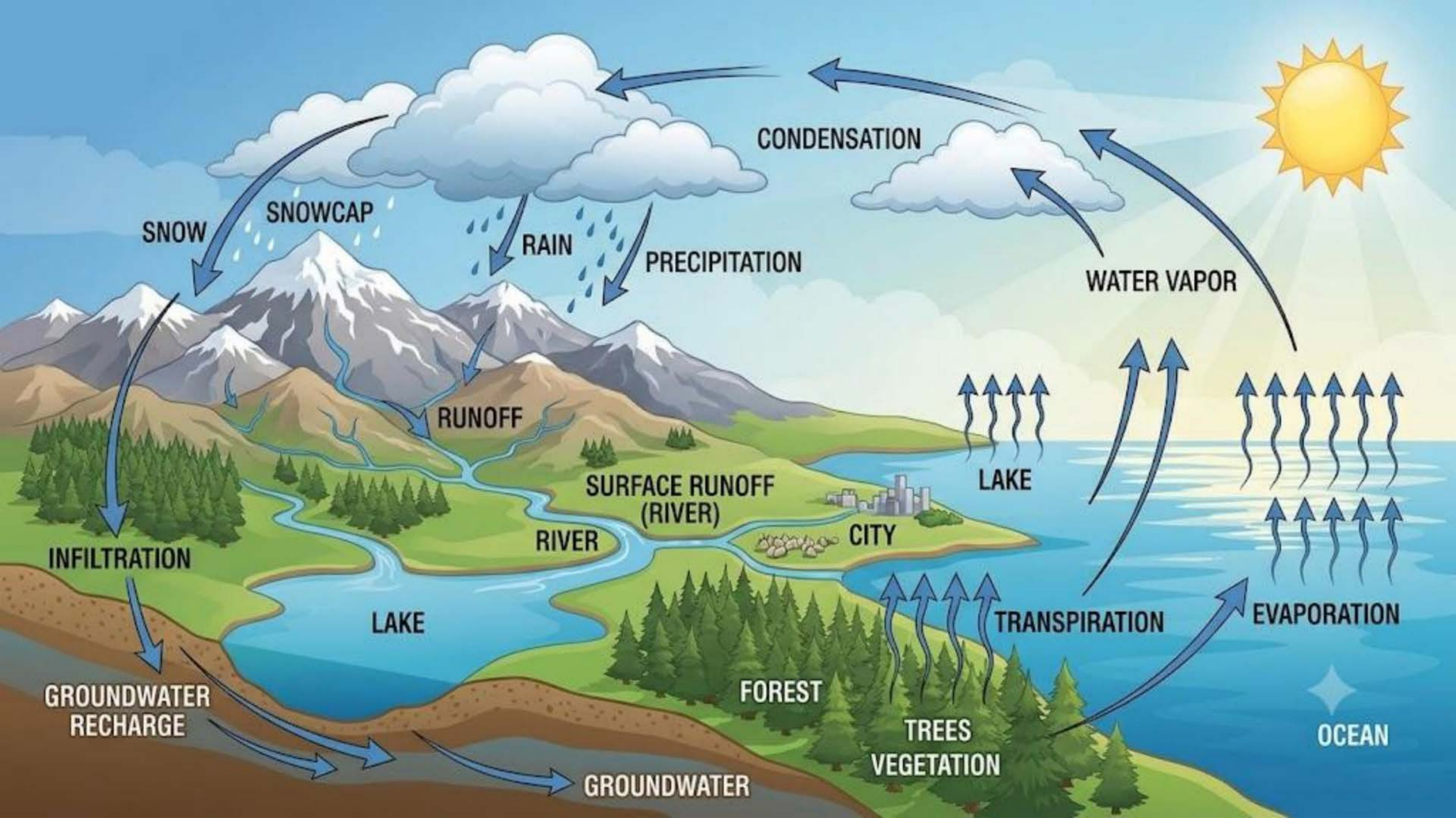
(ii) **Protective Functions:** Forests control the water cycle.

- The thick layer of humus in the forests prevents evaporation of water.
- The humus acts as a natural sponge and helps to soak the rain water in the soil.
- The forest with its complex root system binds the soil thereby preventing soil erosion and loss of nutrients.
- The thick humus over the years is formed by the decay of forest litter, which increases the fertility of the soil.

Importance of Forests

(iii) **Regulatory Functions:** The trees utilise carbon dioxide and release oxygen during photosynthesis.

- The oxygen released by the trees is used by the animal world.
- Thus, the trees perpetuate the cycles of oxygen and carbon dioxide in the biosphere.
- They also regulate the water cycle.
- The trees absorb water from the ground, release water (during transpiration) into the atmosphere which helps to form clouds and precipitation, which brings water again into the soil, thus completing the water cycle.


CONDENSATION
SNOW
SNOWCAP
RAIN
PRECIPITATION
WATER VAPOR
RUNOFF
SURFACE RUNOFF
(RIVER)
LAKE
RIVER
CITY
INFILTRATION
LAKE
TRANSPIRATION
EVAPORATION
FOREST
TREES
VEGETATION
OCEAN
GROUNDWATER
RECHARGE
GROUNDWATER

Importance of Forests

Habitation (Home)

(iv) **Accessory Functions:** Accessory functions of the trees means that the forests provide habitat for the wildlife.

- Forests also provide aesthetics and recreation to human beings through National Parks, Wildlife Sanctuaries and Biosphere Reserves.

Importance of Forests

(v) **Carbon sink:** The Supreme Court of India quoted from a 2009 report of the Ministry of Environment and Forests titled 'India's Forest and Tree Cover: Contribution as a Carbon Sink', saying that India's forests serve as a major sink of carbon dioxide (CO_2).

Absorption of Carbon in the atmosphere
→ Cleaning the air.

The Forest (Conservation) Amendment Bill, 2023 or the Forest Conservation and Augmentation Act

It seeks to amend the Forest Conservation Act, 1980, a legislation enacted to protect India's forests by empowering the Central government to regulate the extraction of forest resources — from timber and bamboo to coal and minerals — by industries as well as forest-dwelling communities.

The new Act seeks to encourage the cultivation of plantations on non-forest land, which can increase tree cover over time, act as a carbon sink of additional 2.5-3.0 billion tons of CO_2 equivalent by 2030 and aid India's ambitions of having 'net zero' carbon emissions by 2070.

It also seeks to create infrastructure that would aid national security and create livelihood opportunities for those residents on the periphery of forests.

Environmental Correlation

Climate Influence:

- The type of vegetation in a region is primarily determined by the climate, particularly rainfall and temperature.
- For instance, tropical rainforests thrive in high rainfall areas, while desert vegetation is adapted to arid conditions.

Soil and Topography:

- Soil type and topography also play significant roles.
- For example, mangrove forests grow in saline coastal soils, while mountain forests vary with altitude.

Environmental Correlation

Biodiversity and Ecological Balance:

- Natural vegetation supports biodiversity, providing habitat for various species.
- It also maintains ecological balance by regulating the water cycle, preventing soil erosion, and contributing to carbon sequestration.
  - ↳ Sinking
  - absorption

Flora → Plant Kingdom

Fauna → Animal Kingdom



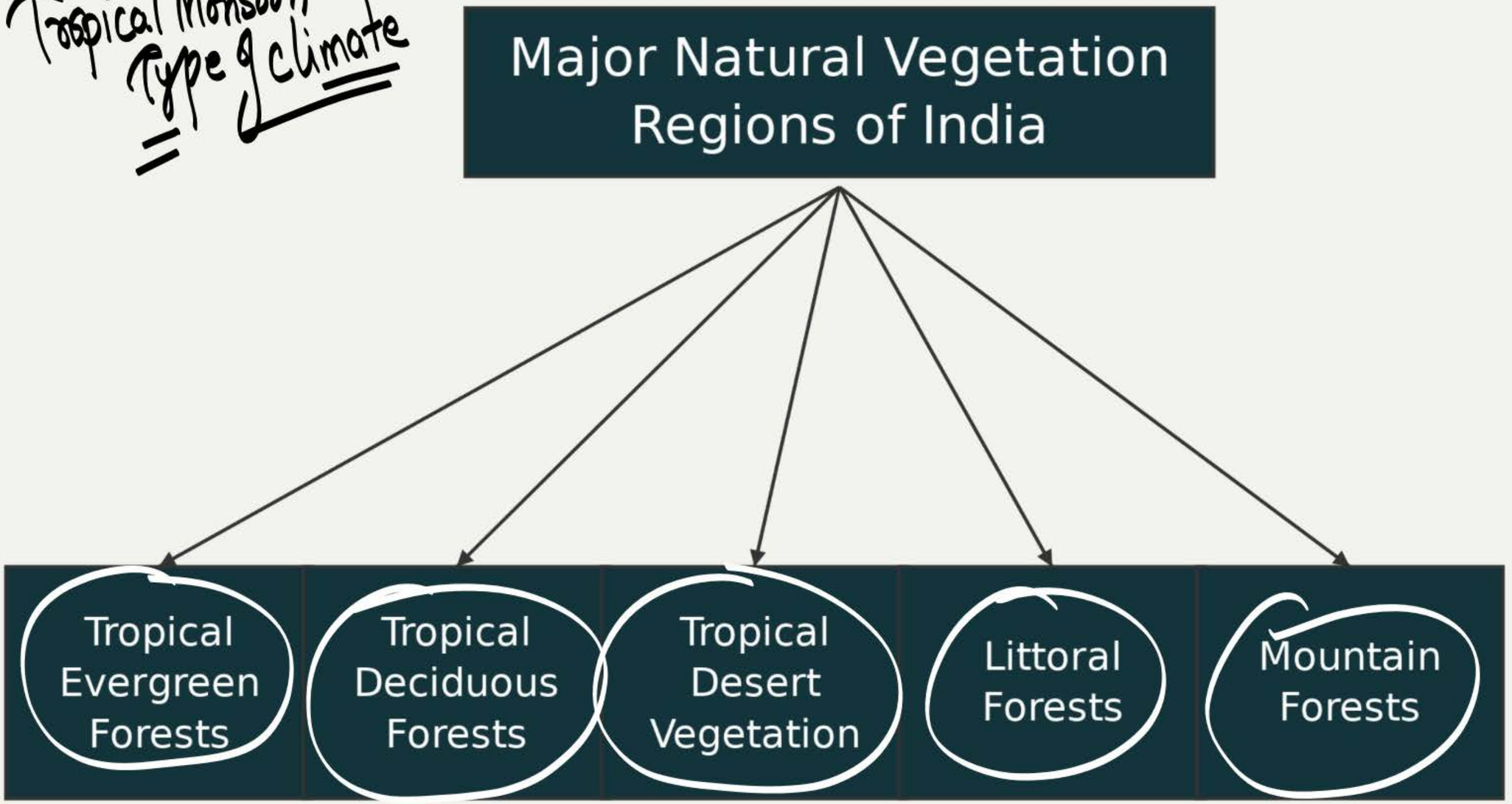
Saturated

Basalt Lava Volcanic & greous

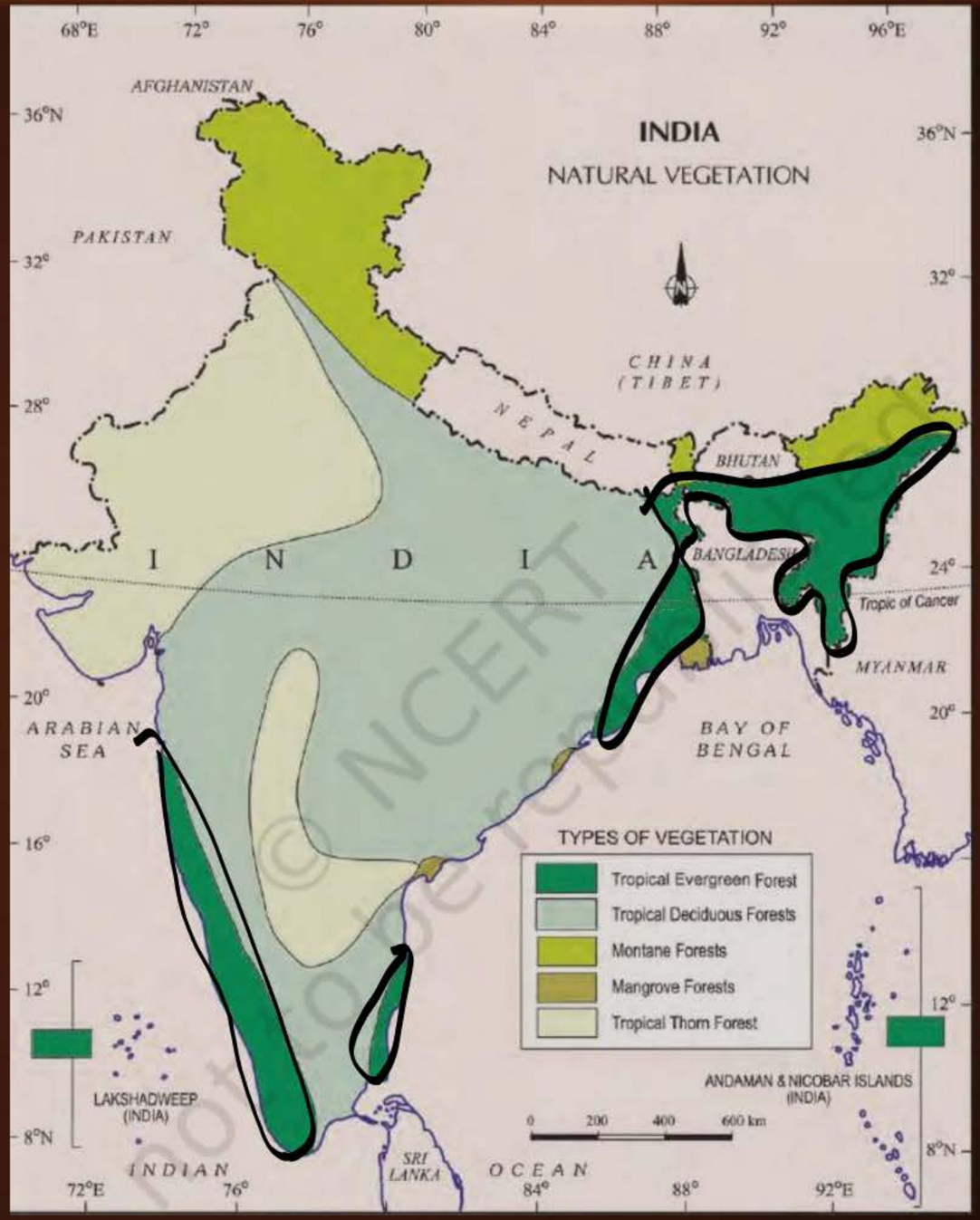
Types of Vegetation

- India has a variety of forests and natural vegetation due to variations in climatic conditions, soil types and relief features.
- **The Western Ghats and the Andaman and Nicobar Islands** are marked with tropical rain forests;
- **the Himalayas** have temperate vegetation;
- **the desert and semi-desert regions of Rajasthan** have a wide variety of bushes and thorny vegetation;
- **the Delta regions** have tropical forests and mangroves.

Tropical Monsoon Type of climate



Ever Green


INDIA
NATURAL VEGETATION
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
TYPES OF VEGETATION
Tropical Evergreen Forest
Tropical Deciduous Forests
Montane Forests
Mangrove Forests
Tropical Thorn Forest
0 200 400 600 km
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N





1. Tropical Evergreen or Rain Forests

(a) **Climatic Conditions:** These forests are found in the areas where

- the annual rainfall is more than 200 cm with a short dry season,
- average annual temperature is between 28°C and 27°c and
- average annual humidity exceeds 77per cent.

1. Tropical Evergreen or Rain Forests

- **Distribution:** These forests are chiefly distributed in the

- **heavy rainfall areas of the western slopes of the Western Ghats,**
- **hills of north-eastern region and the island groups of Lakshadweep,**
- **the Andaman and Nicobar and Tamil Nadu coast.**

1. Tropical Evergreen or Rain Forests

(c) Characteristic Features

(i) Since this region remains warm and wet throughout the year, it has a luxuriant vegetation of all kinds - trees, shrubs and creepers which give it a multilayered structure.

(ii) Trees reach great heights of more than 60 m .

(iii) The carpet layer of herbs and grasses cannot grow because of the dense canopy of trees which do not allow enough sunlight to reach the ground.

(iv) The trees in these forests do not have a fixed time to shed their leaves, to flower or for fruition. That is why these forests appear green all the year round.

Umbrella like Structure

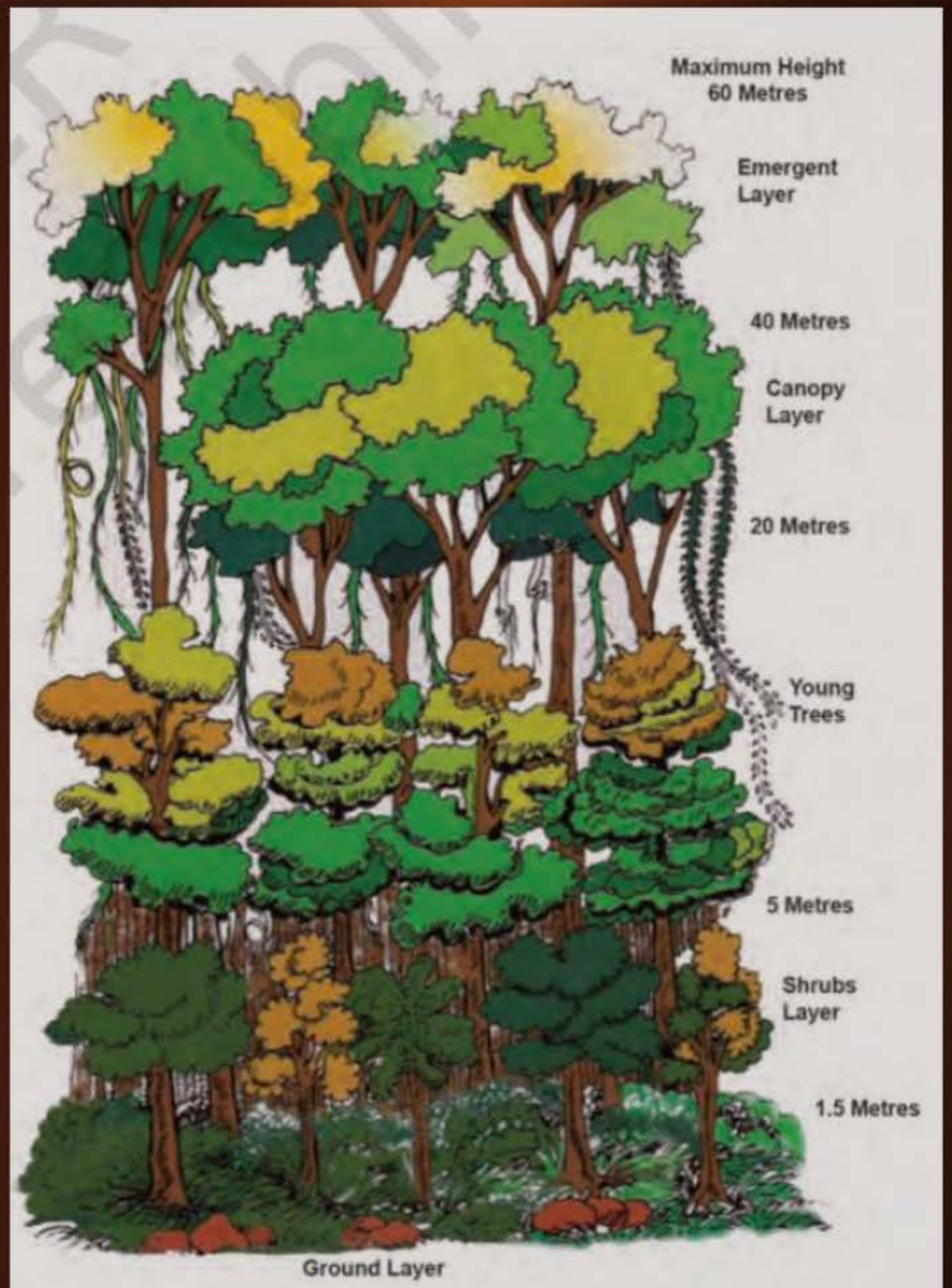


Figure 5.1 : Tropical Evergreen Forest

1. Tropical Evergreen or Rain Forests

The main variety of trees found in these forests include **rosewood, ebony, mahogany, toon, chaplas, sissoo, gurjan, telsur, etc.**


ROSEWOOD
(Dalbergia latifolia)
EBONY
(Diospyros ebenum)
MAHOGANY
(Swietenia macrophylla)
TOON
(Toona ciliata)
CHAPLAS
(Artocarpas chaplasha)
SISSOO
(Dalbergia sissoo)
GURJAN
(Dipterocarpos turbinatus)
(Hollong)
TELSUR
(Hopea odorata)

Types of Vegetation

Problem

① Dense forest → Can't be entered easily.

② Transportation issue

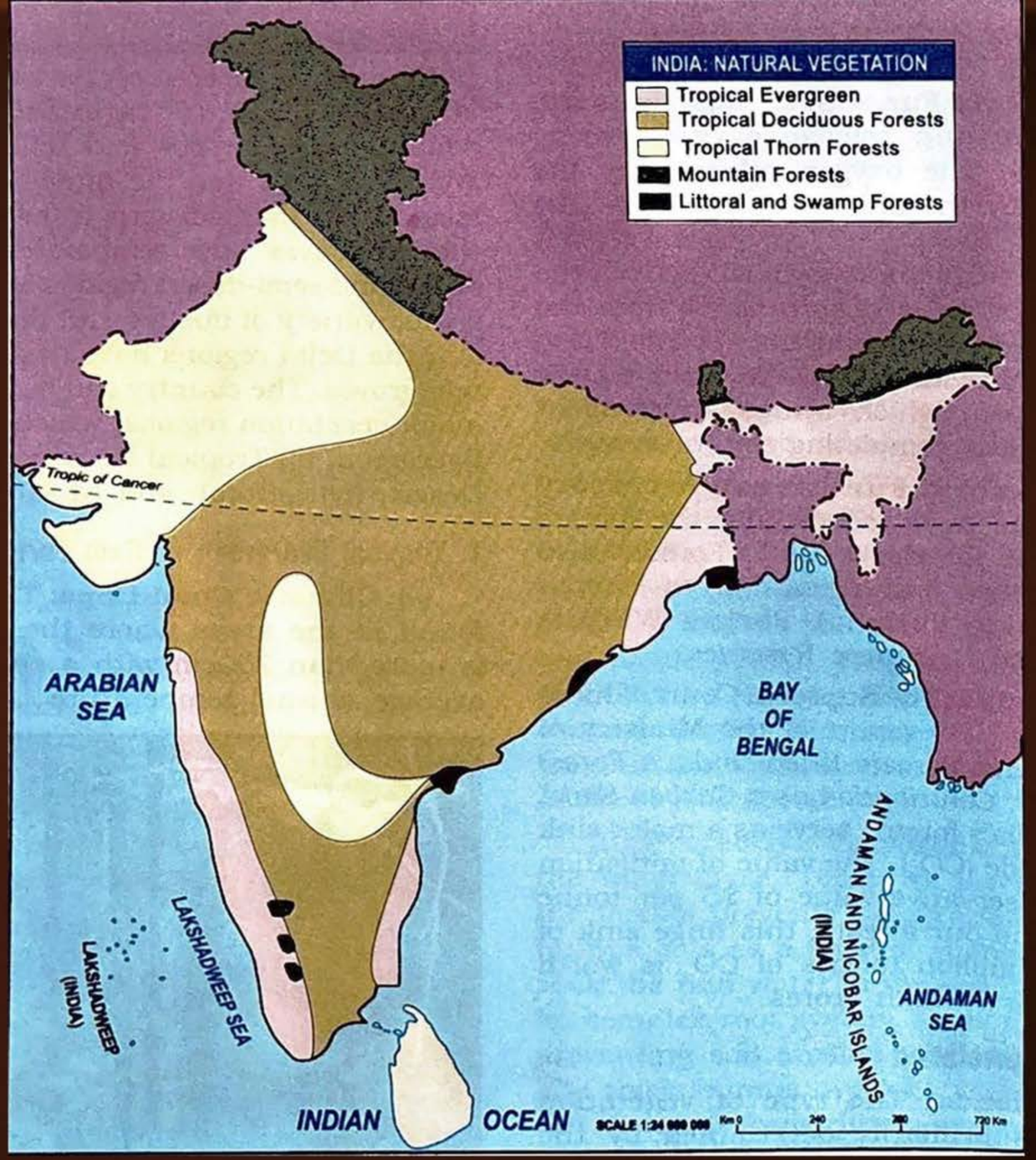
③ Mixed vegetation - plants & trees entangled with each other.

(d) Economical Value: Tropical Evergreen forests produce various plant species of high economic value.

- The timber produced is hard, durable and fine-grained.

(e) Environmental Correlation: These forests maintain high biodiversity and play a crucial role in carbon sequestration.

Natural Vegetation of India





- The type of natural vegetation shown in the image is found in

a. Tropical desert
b. Tropical Deciduous
c. Tropical Evergreen
d. Littoral Forest

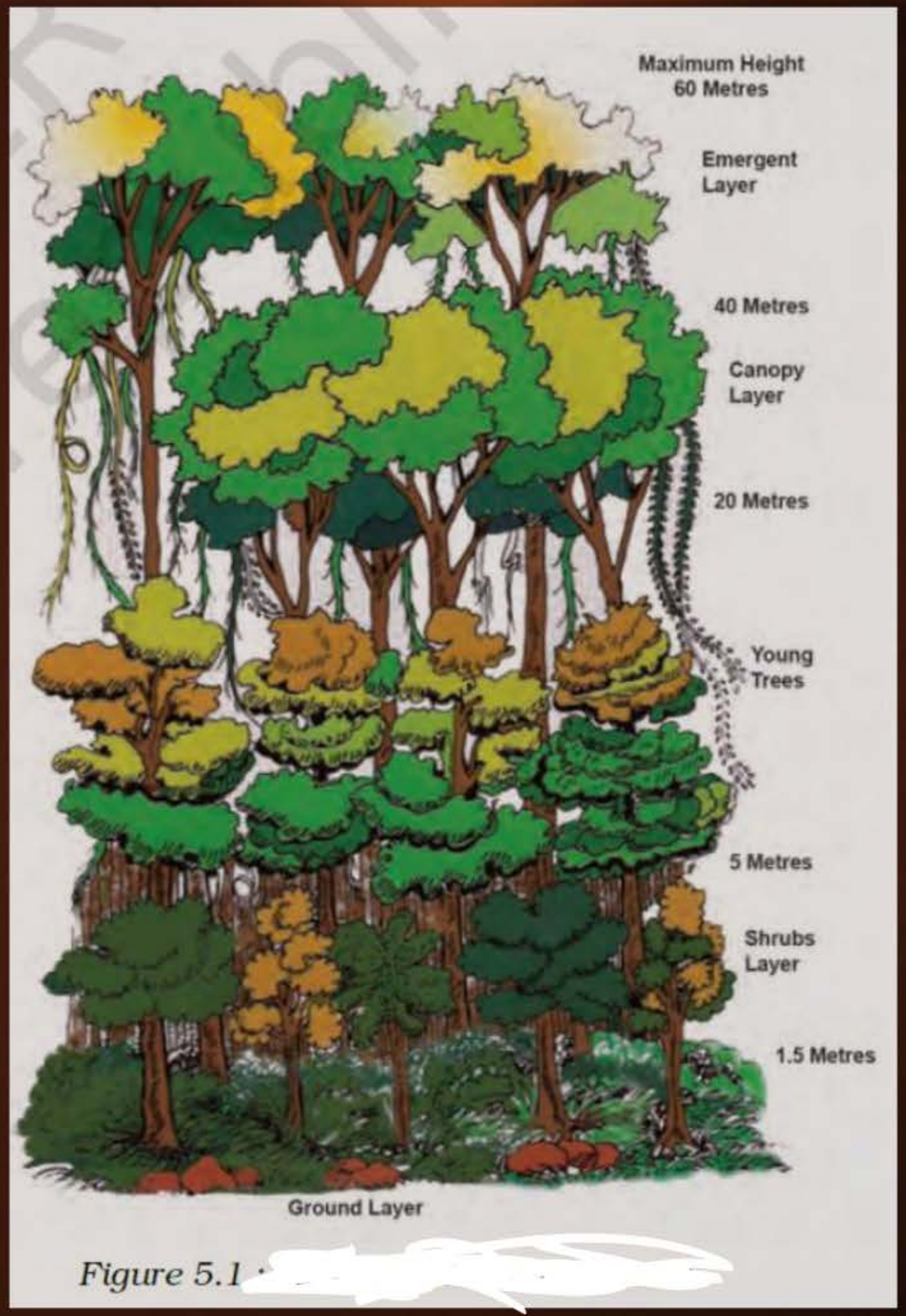


Figure 5.1

Consider the following two statements:

- **Statement 1:** *Trees in Tropical Rain Forests are evergreen and do not shed their leaves.*
- **Statement 2:** *Individual tree species shed their leaves at different times throughout the year due to absence of a distinct dry season.*

A. Both Statements are True, and Statement 2 correctly explains Statement 1.

B. Both Statements are True, but Statement 2 does not explain Statement 1.

C. Statement 1 is False, but Statement 2 is True.

D. Both Statements are False.

PYQ

- Which type of forest is found in the windward side of the western ghats

a. Tropical Evergreen
b. Tropical Deciduous
c. Tropical Desert
d. Littoral Forest



PYQ



a) Name two regions where tropical evergreen forests are found?

b) What are the temperature & rainfall requirements for the growth of these forests?

Thank You